

Study of IP Addressing and Subnetting Date : _____

A Easy level

1. 10.0.0.0/8

→ IP class - A

Default Subnet mask : 255.0.0.0

Network bits : 8

Host bits : 24

usable host per Subnet : $(2^{24}) - 2 = 16,777,214$

Network Address : 10.0.0.0

First usable IP : 10.0.0.1

Last usable IP : 10.255.255.254

Broadcast address : 10.255.255.255

2) 172.16.0.0/16

→ IP class : B

Default Subnet mask : 255.255.0.0

Network bits : 16

Host bits : 16

usable host per Subnet : $(2^{16}) - 2 = 65,534$

Network Address : 172.16.0.0

First usable IP : 172.16.0.1

Last usable IP : 172.16.255.254

Broadcast address : 172.16.255.255

3) 192.168.10.25 / 24

IP class : C

Default Subnet mask : 255.255.255.0

Network bits : 24

Host bits : 8

Usable host Per Subnet : $2^8 - 2 = 254$

Network address : 192.168.10.0

First usable IP : 192.168.10.1

Last usable IP : 192.168.10.254

Broadcast address : 192.168.10.255

4) 199.1.1.1 / 24

IP class : C

Default mask : 255.255.255.0

Network bits : 24

Host bits : 8

Usable host Per Subnet : $2^8 - 2 = 254$

Network address : 199.1.1.0

First usable IP : 199.1.1.1

Last usable IP : 199.1.1.254

Broadcast address : 199.1.1.255

5) IP Address IP class Default Subnet Mask

15.10.5.20	A	255.0.0.0
175.20.40.10	B	255.255.0.0
210.100.5.1	C	255.255.255.0

6) 192.168.1.100/24

Network Address : 192.168.1.0
 First usable IP : 192.168.1.1
 Last usable IP : 192.168.1.254
 Broadcast address : 192.168.1.255

7) 172.20.5.0/24

First usable IP : 172.20.5.1
 Last usable IP : 172.20.5.254

8) IP Address IP class

5.1.2.3	A
130.20.1.1	B
192.168.1.1	C
230.10.10.1	D
250.1.1.1	E

9) 192.168.50.10/25

Network address: 192.168.50.0

Broadcast address: 192.168.50.127

First usable IP: 192.168.50.1

Last usable IP: 192.168.50.126

Usable host Per Subnet: $(2^7 - 2) = 126$

10) 10.20.30.40/16

Network address: 10.20.0.0

Broadcast address: 10.20.255.255

First usable IP: 10.20.0.1

Last usable IP: 10.20.255.254

Usable host Per Subnet: $(2^{16} - 2) = 65534$

8 Medium Level

1) 200.100.50.75 / 26

Class : C

Default Subnet mask : 255.255.255.0

Subnet mask : 255.255.255.192 (126)

Network bits : 26

Host bits : $32 - 26 = 6$

Network address = 200.100.50.64

First usable IP = 200.100.50.65

Last usable IP = 200.100.50.126

Broadcast address = 200.100.50.127

2) 130.4.102.1 / 22

Class : B

Default Subnet mask : 255.255.0.0

Subnet mask : 255.255.252.0 (122)

Network bits : 22

Host bits : 10

Network Address : 130.4.100.0

First usable IP : 130.4.100.1

Last usable IP : 130.4.103.254

Broadcast address : 130.4.103.255

37) 203.1.1.0 / 28

Class : C

Default class C mask = /24

Given mask = /28

Borrowed bits = $28 - 24 = 4$ borrowed bits.

Subnet mask : 255.255.255.240 (/28)

Number of Subnets : $2^4 = 16$ Host bits : $32 - 28 = 4$ Usable hosts per subnet : $2^4 - 2 = 14$

First four Subnet blocks :

Subnet	Network Address	Usable IP Range	Broadcast
1	203.1.1.0	203.1.1.10 - 203.1.1.14	203.1.1.15
2	203.1.1.16	203.1.1.17 - 203.1.1.30	203.1.1.31
3	203.1.1.32	203.1.1.33 - 203.1.1.46	203.1.1.47
4	203.1.1.48	203.1.1.49 - 203.1.1.62	203.1.1.63

4) 216.21.5.0 / 24

Requirement = 30 usable hosts per Subnet

Formula : $2^4 - 2 \geq 30$

$\therefore 2^5 - 2 = 30$, We need 5 host bits

IPv4 has 32 bits : $32 - 5 = /27$

$\therefore$ New Subnet mask = 255.255.255.224

Borrowed bits

original network 4 New network.
(/24) (/27)

$\therefore 27 - 24 = 3$ borrowed bits

→ Number of Subnets = $2^3 = 8$

Usable hosts per Subnet : 30

New Subnet mask = 255.255.255.224 (/27)

First two Subnet blocks :

Subnet	Network Address	usable IP Range	Broadcast
1	216.21.5.0	216.21.5.1 - 216.21.5.30	216.21.5.31
2	216.21.5.32	216.21.5.33 - 216.21.5.62	216.21.5.63

→ First Subnet : 216.21.5.0 / 27

Second Subnet : 216.21.5.32 / 27

5) 192.168.100.0/26

Borrowed bits = $(26 - 24) = 2$

Number of Subnets = $2^2 = 4$

Host bits = $32 - 26 = 6$

Usable hosts per subnet = $2^6 - 2 = 62$

Subnet mask : 255.255.255.192 (126)

First three Subnet blocks :

Subnet	Network Address	usable IP range	Broadcast
1	192.168.100.0	192.168.100.1 - 192.168.100.62	192.168.100.63
2	192.168.100.64	192.168.100.65 - 192.168.100.126	192.168.100.127
3	192.168.100.128	192.168.100.129 - 192.168.100.190	192.168.100.191

6) 172.16.17.30/20 and 172.16.28.15/20

→ Here /20 means:

Subnet mask : 255.255.240.0

Block size in the 3rd Octet :

$$255 - 240 = 16$$

∴ Subnet ranges : 0-15, 16-31, 32-47
Both 17 and 28 fall in the 16-31 range.

Yes they are in the same Subnet and both have the network address as 172.16.16.0/20

7) 182.44.82.16 /26

Network address : 182.44.82.0

Board address : 182.44.82.63

Usable host range : 182.44.82.1 -

182.44.82.62

8) 192.168.10.25/23 and 192.168.11.200/23

→ Here /23 means

Subnet mask : 255.255.254.0

Block Size in the 3rd Octet : $256 - 254 = 2$

∴ Subnet ranges : 0-1, 2-3, 4-5, ..., 10-11, 12-13

1P1: 192.168.10.25/23

Third Octet : 10

(10-11) Subnet starts at 10

Network address : 192.168.10.0

IP2 : 192.168.11.200 / 23

Third octet : 11

C10-11) Subnet range

Network address : 192.168.10.0

→ Yes, they are in the same Subnet
and both have the network address
as 192.168.10.0 / 23

9) 172.16.10.100 / 20

Subnet mask - 255.255.240.0

usable hosts per Subnet - $(2^{12} - 2) = 4094$

Network address - 172.16.0.0

First usable IP - 172.16.0.1

Last usable IP - 172.16.15.254

10) 192.168.10.0 / 24

Requirements = Atleast 14 Subnets

$2^n \geq 14$ $2^4 = 16$, 4 bits must be borrowed

Minimum borrowed bits - 4

New Subnet mask - 255.255.255.240

Number of Subnets created : 16 (2^4)

Usable hosts per Subnet : 14 ($2^4 - 2$)

C hand level

1) 10.10.0.0/16

→ Requirement : Atleast 500 usable hosts per Subnet

Minimum prefix length : /23

Subnet mask : 255.255.254.0

Borrowed bits : 7 bits

usable hosts per Subnet : $(2^7) - 2 = 126$

number of equal Subnets : $2^7 = 128$

First four subnet blocks

Subnet	Network	usable range	Broadcast
1	10.10.0.0/23	10.10.0.1 - 10.10.1.254	10.10.1.255
2	10.10.2.0/23	10.10.2.1 - 10.10.3.254	10.10.3.255
3	10.10.4.0/23	10.10.4.1 - 10.10.5.254	10.10.5.255
4	10.10.6.0/23	10.10.6.1 - 10.10.7.254	10.10.7.255

2) 172.20.0.0/16

Requirement : Atleast 1000 usable hosts
Per subnet.

Minimum prefix length : /22

Subnet mask : 255.255.252.0

Borrowed bits : 6 bits

Number of Subnets : $2^6 = 64$

First four Subnet addresses:

Subnet addresses

Broadcast addresses

172.20.0.0/22

172.20.3.255

172.20.4.0/22

172.20.7.255

172.20.8.0/22

172.20.11.255

172.20.12.0/22

172.20.15.255

3) 150.10.0.0/16

Requirement : Atleast 100 departments,
each requiring 250 usable hosts.

Minimum subnet size : 256 addresses

Prefix : /24

Subnet mask : 255.255.255.0

Borrowed bits : 3 bits.

Number of available Subnets : $2^8 = 256$

unused Subnets : $256 - 100 = 156$

usable hosts per Subnet : 254

first four Subnet addresses :

- i) $170.10.0.0 / 24$
- ii) $170.10.1.0 / 24$
- iii) $170.10.2.0 / 24$
- iv) $170.10.3.0 / 24$

4) $172.31.100.0 / 21$

→ $/21$ has a block size of $256 - 248 = 8$
 So 3rd octet boundaries are :
 $0, 8, 16, 24, \dots, 96, 104 \dots$

Network address : $172.31.96.0 / 21$

Broadcast address : $172.31.103.255$

usable range : $172.31.96.1 \rightarrow 172.31.103.254$

usable hosts per Subnet : 2046

Number of $/21$ Subnets obtainable

$$: (2^{21-16}) = 32$$

from the original class B / 16 network

5) 11.0.0.0/8

Requirement : 2046 usable hosts each

Host bits : $(2^n - 2) = 2046 \therefore 11$

Minimum prefix : /21

Subnet mask : 255.255.248.0

Borrowed bits : 13 bits

Number of Subnets : $2^{13} = 8192$

first four Subnet addresses

i) 11.0.0.0/21

ii) 11.0.8.0/21

iii) 11.0.16.0/21

iv) 11.0.24.0/21

6) 100.10.0.0/8

Requirement : Atleast 1200 usable hosts
Per Subnet.

Minimum Prefix : /21

Subnet mask : 255.255.248.0

Borrowed bits : 13 bits

Usable hosts per subnet : 2046

Number of equal Subnets : 8192

first five Subnet addresses:

- i) 100.0.0.0 / 21
- ii) 100.0.8.0 / 21
- iii) 100.0.16.0 / 21
- iv) 100.0.24.0 / 21
- v) 100.0.32.0 / 21

7) 192.168.40.0 / 24

Requirements : Department A → 100 hosts
Department B → 50 hosts
Department C → 25 hosts

using VLSM :

→ Department A - 100 hosts
It needs 126 usable : $2^7 - 2 = 126$

Prefix : /25

Subnet mask : 255.255.255.128

Network address : 192.168.40.0 /25

Usable range : 192.168.40.1 -
192.168.40.126

Broadcast address : 192.168.40.127

→ Department B - 50 hosts
It needs 62 usable : $2^6 - 2 = 62$

Prefix : /26

Subnet mask : 255.255.255.192

Network address : 192.168.40.128 / 26

usable range : 192.168.40.129 -

192.168.40.190

Broadcast address : 192.168.40.191

→ Department c - 25 hosts

It needs 30 usable : $2^5 - 2 = 30$

Prefix : /27

Subnet mask : 255.255.255.224

Network address : 192.168.40.192 / 27

usable range : 192.168.40.193 -

192.168.40.222

Broadcast address : 192.168.40.223

8) 172.16.0.0 / 16

Requirement : 400 hosts

200 hosts

100 hosts

50 hosts

→ 400 hosts

Needs $(2^9 - 2) = 510$ usable

Prefix : /23

Network address : 172.16.0.0/23

usable range : 172.16.0.1 -
172.16.1.254

Broadcast address : 172.16.1.255

→ 200 hosts

Needs $(2^8 - 2) = 254$ usable

Prefix : /24

Network address : 172.16.2.0/24

usable range : 172.16.2.1 -
172.16.2.254

Broadcast address : 172.16.2.255

→ 100 hosts

Needs = 126 usable

Prefix : /25

Network address : 172.16.3.0/25

usable range : 172.16.3.1 -
172.16.3.126

Broadcast address : 172.16.3.127

→ 50 hosts

Needs = 62 usable.

Prefix : 126

Network address : 172.16.3.128 / 26

Usable range : 172.16.3.129 -
172.16.3.190

Broadcast address : 172.16.3.191

9) Host A : 172.16.70.130 / 18

Host B : 172.16.130.200 / 18

Here /18 mask is 255.255.192.0

The 3rd octet block size : $256 - 192 = 64$

∴ Subnet ranges :

0 - 63, 64 - 127, 128 - 191, 192 - 255

Host A

Host B

Network : 172.16.64.0 / 18

Network : 172.16.128.0 / 18

Usable range : 172.16.64.1

Usable range : 172.16.128.1

172.16.127.254

172.16.191.254

Broadcast : 172.16.127.255

Broadcast : 172.16.191.255

→ No, the two hosts are not in the same Subnet.

10) 192.168.1.0/24

Requirements : Atleast 6 equal size
Subnets Each Subnetting atleast
30 usable hosts

Needs atleast 30 : $(2^5 - 2) = 30$
∴ we need 5 host bits

Fixed-length Subnetting possible = Yes
Prefix : /27

Subnet mask : 255.255.255.254

number of Subnets = $2^3 = 8$

usable hosts per Subnets : $(2^5 - 2) = 30$

All Subnet blocks :

Network	Usable Range	Broadcast
192.168.1.0/27	192.168.1.1 - 192.168.1.30	192.168.1.31
192.168.1.32/27	192.168.1.33 - 192.168.1.62	192.168.1.63
192.168.1.64/27	192.168.1.65 - 192.168.1.94	192.168.1.95
192.168.1.96/27	192.168.1.97 - 192.168.1.126	192.168.1.127

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192.168.1.128/27

192.168.1.129 -

192.168.1.159

192.168.1.158

192.168.1.160/27

192.168.1.161 -

192.168.1.191

192.168.1.190

192.168.1.192/27

192.168.1.193 -

192.168.1.223

192.168.1.222

192.168.1.224/27

192.168.1.225 -

192.168.1.255

192.168.1.254